

EMPOWERING SERVICE OPERATIONS WITH REAL-TIME PRODUCTION VISIBILITY

A Comprehensive White Paper on the Daily RO Production Dashboard
Read-Only DB2 Integration for Hours per Repair Order, Hourly Monitoring,
and Day-over-Day Performance Comparison

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Prepared for Service Operations, Fleet Maintenance & Repair Organizations

Executive Highlights

- **Problem:** Manual, delayed, and fragmented visibility into daily labor hours per Repair Order (RO) and productivity patterns.
- **Solution:** Lightweight, secure, browser-based Streamlit dashboard connected to DB2 via read-only API.
- **Impact:** Instant daily production reports, hourly monitoring, automated day-over-day comparison, and one-click exports — all without modifying source data.

Table of Contents

1. Executive Summary
2. The Operational Challenge
3. The Solution: Daily RO Production Dashboard
4. Technical Architecture & Security Model
5. Core Features & User Experience
6. Implementation & Integration Guide
7. Business Value & Return on Investment
8. Recommended Use Cases
9. Conclusion & Recommended Next Steps

1. Executive Summary

Service, repair, and maintenance organizations live and die by their ability to understand daily output. How many Repair Orders (ROs) were worked on today? How many hours were actually logged versus planned? Did productivity peak in the morning or afternoon? How does today compare to yesterday or last week?

Traditional approaches — pulling data manually from DB2 into spreadsheets, running overnight batch reports, or relying on tribal knowledge — create lag, errors, and missed opportunities. Front-line supervisors and operations managers need **same-day, granular, visual insights** without burdening IT or risking data integrity.

The Daily RO Production Dashboard solves this by delivering a purpose-built, read-only web interface that connects directly to your DB2 environment through a secure API layer. In minutes, users can:

- View total logged hours, number of active ROs, and average hours per RO for any selected day
- Drill into detailed hours-per-RO breakdowns with technician and department attribution
- Analyze hourly production patterns to identify peak productivity windows and workflow bottlenecks
- Compare any two days side-by-side with automatic variance calculation and visual hourly overlays
- Export professional multi-sheet Excel reports or CSV extracts with one click

Key Differentiator: The entire solution is designed for **read-only** access. No risk to production data. No complex ETL. No new database permissions. Just secure, fast visibility that turns raw DB2 labor records into actionable operational intelligence.

2. The Operational Challenge

In heavy equipment, truck, automotive, and fleet service environments, labor is the largest controllable cost and the primary driver of throughput. Yet most organizations suffer from the same visibility gaps:

Delayed Insights: Daily production numbers often arrive 24–48 hours late via overnight reports or manual Excel consolidation.

Granularity Loss: Aggregated totals hide critical details — which ROs consumed disproportionate hours? Which technicians or bays were bottlenecks?

No Temporal Pattern Analysis: Without hourly breakdowns, managers cannot see whether work starts late, ends early, or has productivity dips after lunch.

Comparison Friction: Day-over-day or week-over-week analysis requires manual data wrangling, leading to inconsistent metrics and version control issues.

IT Dependency: Every new report or filter request goes through IT, creating backlog and frustration.

Data Security Concerns: Giving broad query access or exporting large datasets increases risk of accidental or malicious data exposure.

These challenges compound in high-volume operations where hundreds of labor entries are logged daily across multiple bays, shifts, and technicians. The result is reactive management instead of proactive optimization.

3. The Solution: Daily RO Production Dashboard

The **Daily RO Production Dashboard** is a modern, browser-based application built with Streamlit that provides immediate, self-service access to production metrics derived from your existing DB2 labor data.

3.1 Core Value Proposition

It transforms raw time entries (typically one row per labor line on a Repair Order) into four high-value views. **v2.0 adds** a modern HTML frontend option, Weekly Production Tracker (matching your existing Excel format with per-day Earned/Goal columns and Hours To Goal), Efficiency metric ($\text{Actual Hours} \div \text{Unique Techs} \times 7 \text{ expected man-hours/day}$), and interactive 7/30-day trend charts.

1. Daily Production Summary — Instant KPIs for any chosen date: Total Logged Hours, Unique ROs worked, Average Hours per RO, with optional comparison to a benchmark day.

2. Hours per RO Detail — Sortable, filterable table showing every Repair Order, the technician(s) who worked on it, department/bay, total hours logged, number of time entries, and percentage of the day's total labor.

3. Hourly Monitoring — Visual breakdown of when work was performed throughout the day. Bar charts reveal peak productivity windows; line chart overlays allow direct comparison of work patterns between two different days.

4. Day-over-Day Comparison — Side-by-side metrics with automatic delta and percentage change calculations, plus interpretation guidance for operations leaders.

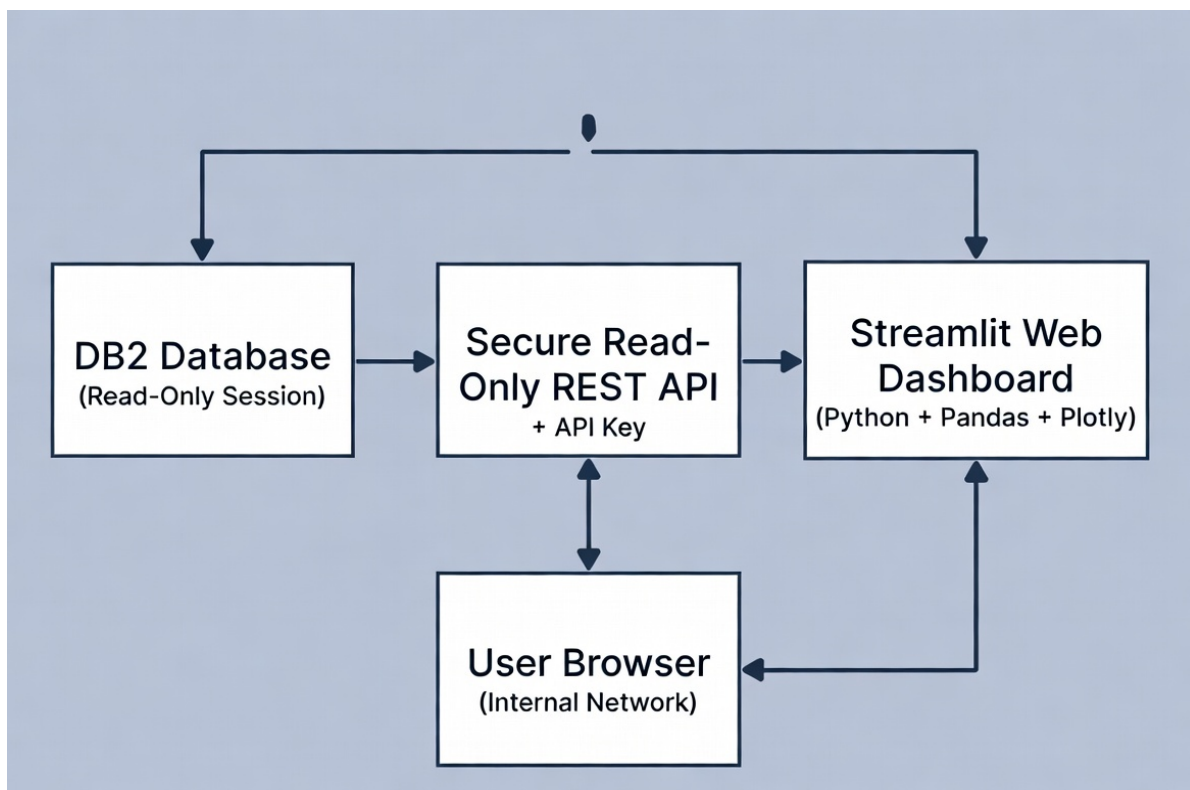


Figure 1: High-level technical architecture — secure read-only data flow from DB2 to end users

4. Technical Architecture & Security Model

The solution follows a clean, minimal, and highly secure architecture that respects existing DB2 governance:

4.1 Data Flow

1. **DB2 Source** — Existing labor detail tables (read-only session or service account).
2. **API Layer** — Your organization's existing read-only REST API / data gateway (or a lightweight new endpoint). The dashboard never connects directly to DB2.
3. **Streamlit Application** — Python-based web app running on an internal server or container. All heavy lifting (aggregation, filtering, charting) happens server-side.
4. **Browser** — Zero-install access for authorized users on the internal network.

4.2 Security & Compliance Advantages

- Uses your existing read-only API key / token — no new DB2 credentials distributed to end users
- All queries are read-only by design; the application contains no INSERT, UPDATE, or DELETE capability
- API key can be scoped to specific views or date ranges if your gateway supports it
- No data is stored locally in the dashboard beyond session cache (default 5-minute TTL)
- Supports internal network deployment behind VPN or SSO if required
- Full audit trail possible by logging API calls at the gateway layer

Important: Because the dashboard is read-only and relies on your existing API governance, it typically requires minimal security review compared to tools that require direct database access or write privileges.

5. Core Features & User Experience

The interface was designed for speed and clarity — operations managers should be able to open the tool, select a date, and immediately understand the story of the day.

5.1 Main Dashboard View

Upon generating a report, users see four prominent KPI cards at the top, followed by tabbed detailed views. Filters for Technician and Department appear dynamically once data is loaded, allowing focused analysis (e.g., "Show me only Bay 2 performance today").

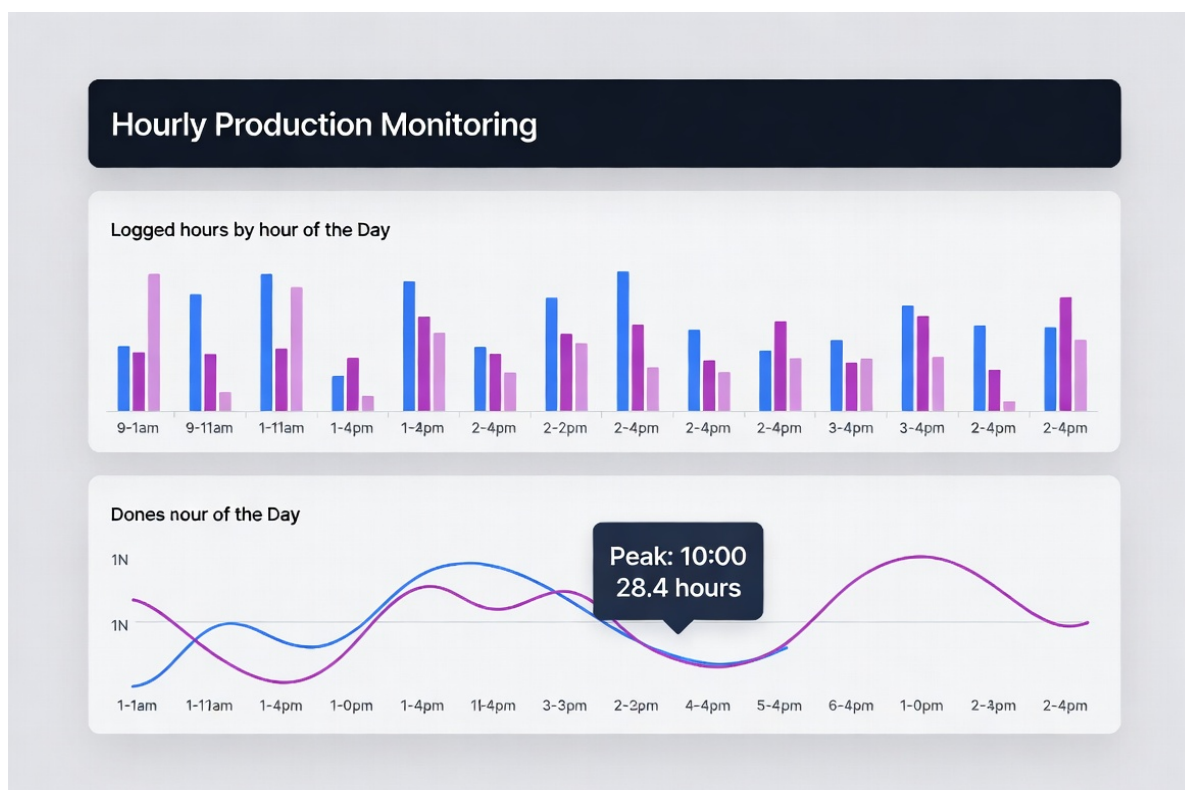


Figure 2: Hourly Production Monitoring view with bar chart and day-over-day line overlay

5.2 Hourly Monitoring & Pattern Analysis

One of the most powerful and unique capabilities is the ability to see **when** work happened, not just how much. The hourly bar chart reveals natural rhythms of the shop. The comparison line chart (shown when a benchmark day is selected) makes it immediately obvious whether today started slower, had an earlier lunch, or finished stronger than the comparison day.

5.3 Export & Sharing

Every view can be exported. The multi-sheet Excel export includes an Executive Summary with all KPIs and deltas, plus separate sheets for the RO-level detail, hourly breakdown, and raw labor logs. This makes it trivial to attach a professional report to daily stand-up emails or archive for historical analysis.

6. Implementation & Integration Guide

The dashboard is delivered as a complete, ready-to-run Python project. Typical implementation timeline for organizations with an existing read-only API is **1–3 days**.

6.1 Quick Start (Demo Mode)

The application ships with a high-fidelity mock data generator. Operations leaders and stakeholders can immediately explore the full feature set without any database connection:

1. Clone or copy the project folder
2. `pip install -r requirements.txt`
3. `streamlit run app.py`
4. Toggle "Use Mock Data (Demo Mode)" — ON by default

6.2 Connecting to Your Real DB2 Environment

Once you have API documentation and a read-only key:

1. Store the API key securely in `.streamlit/secrets.toml` (never committed to source control).
2. Edit the `fetch_data()` function in `app.py` to call your specific endpoint and map response fields to the expected DataFrame columns (`ro_number`, `technician`, `logged_hours`, `log_timestamp` or `hour`, `department`).
3. Set `USE MOCK DEFAULT = False`.
4. Test with known good dates that contain labor data.
5. Deploy to an internal web server or container (Docker, Kubernetes, or simple VM) with appropriate network access to your API gateway.

6.3 Customization Points

Common easy extensions include adding 'Expected Hours' or 'Billed Hours' columns (if available in your DB2 schema) to calculate efficiency percentages, adding a 7- or 30-day trend tab, implementing role-based default filters, or scheduling daily PDF/email reports using external orchestration.

7. Business Value & Return on Investment

While the primary goal is operational visibility, the downstream business impacts are significant:

7.1 Quantifiable Benefits

Reduced Manual Reporting Time: Supervisors and analysts who previously spent 30–90 minutes per day consolidating data can now generate a full professional report in under 60 seconds.

Faster Issue Detection: Low-output days, unusual RO hour concentrations, or abnormal hourly patterns become visible the same day instead of days or weeks later.

Improved Resource Planning: Understanding true hourly rhythms helps with staffing, break scheduling, and bay utilization decisions.

Better Coaching & Accountability: Technician- and RO-level detail enables targeted conversations grounded in data rather than anecdote.

Reduced Data Risk: Because access is mediated through a governed read-only API and the tool itself cannot modify data, the risk surface is dramatically smaller than broad SQL access or frequent large exports.

7.2 Strategic Value

Beyond daily operations, the dashboard creates a foundation for continuous improvement programs. Consistent, trusted daily metrics are a prerequisite for meaningful lean initiatives, performance benchmarking across locations, and data-driven capacity planning. It also reduces the "shadow IT" problem where well-intentioned managers build their own fragile spreadsheets.

8. Recommended Use Cases

Daily Morning Stand-up (15 minutes)

Service Manager opens the dashboard for yesterday's date, reviews total output vs plan, identifies any outlier ROs or technicians, and shares the hourly pattern insight with the team.

Weekly Performance Review

Compare Monday vs same day last week, or this week vs last week aggregate. Use the comparison view to discuss what changed in workflow or volume.

Technician Coaching

Filter to a specific technician on a low-output day. Review which ROs took longer than average and discuss root causes (parts, complexity, training needs).

Capacity & Scheduling Decisions

Analyze peak hours across multiple days to determine optimal shift start times or whether additional bay capacity is needed during certain windows.

Audit & Compliance Support

When questions arise about labor allocation on specific ROs, the detailed export provides a defensible, timestamped record without requiring IT to run custom queries.

9. Conclusion & Recommended Next Steps

The Daily RO Production Dashboard represents a high-impact, low-risk way to bring modern self-service analytics to service and repair operations that are still heavily dependent on DB2 as a system of record. By leveraging existing read-only API infrastructure and a lightweight Python web framework, organizations can deliver immediate value to front-line leaders without new infrastructure projects or data governance headaches.

Recommended Next Steps

1. Schedule a 30-minute demo with key operations stakeholders using the included mock data mode.
2. Identify the owner of your read-only DB2 API / data gateway and request the necessary documentation and test credentials.
3. Pilot the tool with one high-volume location or department for 2–4 weeks.
4. Gather feedback on the most valuable additional filters, metrics, or export formats.
5. Plan a phased rollout with training focused on interpretation of the hourly patterns and variance signals.

Bottom Line: In an environment where every labor hour matters, the ability to see exactly how those hours were spent — today, not tomorrow — is a powerful competitive advantage. This dashboard delivers that visibility safely, quickly, and beautifully.

For questions, customization requests, or assistance integrating with your specific DB2 API, contact your internal analytics or IT service team. The solution is fully open for extension and owned by your organization.